INCH-POUND
MIL-DTL-32369 (OS)
27 April 2011

DETAIL SPECIFICATION

CARTRIDGE, IMPULSE, CCU-167/B

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document outlines the requirements for the manufacture, assembly, and packaging of the Impulse Cartridge, CCU-167/B and the verification methods upon which product acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents specified in section 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1. Specification, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-D-21625 Design and Evaluation of Cartridges for Cartridge Actuated Devices

MIL-DTL-83420 Wire Rope, Flexible, for Aircraft Control, General Specification for

Comments, suggestions, or questions on this document should be addressed to Commander, Indian Head Division, Naval Surface Warfare Center, Technical Information Branch (E143D), 101 Strauss Avenue, Indian Head, MD 20640 5035, or emailed to amanda.penn@navy.mil. Since contact information can change, you may want to verify the currency of this information using the ASSIST Online database at **https://assist.daps.dla.mil**..

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-810	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-129	Marking For Shipment and Storage
MIL-STD-1168	Ammunition Lot Numbering and Ammunition Data Card
DOD-STD-2101	Classification of Characteristics

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

DRAWINGS

Naval Air Systems Command (Cage Code 14083)

3205AS372	Cartridge, Impulse, CCU-167/B
3205AS369	Case and Bridgewire Assembly

PUBLICATIONS

Naval Sea Systems Command (Cage Code 53711)

OP 5	Ammunition and Explosives Ashore

(Application for copies should be addressed to the DEPARTMENT OF THE NAVY, Indian Head Division, NSWC, Code E143H, Document Control, 4072 North Jackson Road Suite 106, Indian Head, MD 20640-5115 OFFICIAL BUSINESS.)

2.3 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E 545	Image Quality in Direct Thermal Neutron Radiographic Examination
ASTM-E1742/E1742M	Standard Practice for Radiographic Inspection

(Copies of these documents are available online at http://www.astm.org/ or from the American Society for Testing and Materials Customer Service, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

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AMERICAN NATIONAL STANDARD INSTITUTE (ANSI)/AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 Sampling Procedures and Tables for Inspection by Attributes

(Copies of these documents are available online at http://webstore.ansi.org/ansidocstore/default.asp, or from the American Society for Quality, PO Box 3005, 600 N. Plankinton Ave., Milwaukee, WI 53203)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2) in the contract or purchase order a representative sample of impulse cartridges shall be subjected to first article inspection in accordance with 4.1. A first article inspection shall be required if two years or more have passed since the item was manufactured, major turn over in personnel, major changes to the production facility or transfer of production to a new facility or site.

3.2 Serial number. Each impulse cartridge shall be identified with a unique serial number.

3.3 Flight critical parts. This impulse cartridge is a flight critical part relating to personnel safety and emergency egress and shall be identified and procured as such in accordance with NAVAIRINST 4200.25.

3.4 Design and construction. The impulse cartridge shall be of the simplest and most efficient design consistent with the proposed use, and shall meet the requirements of NAVAIR Drawing 3205AS372 and all requirements established herein.

3.4.1 Product baseline. The manufacturer shall maintain a complete product baseline list, specifying each item used in the impulse cartridge, which shall contain description, part number, or identifying number, revision letter and material and/or applicable specification. This list and subsequent revisions shall be submitted for technical review and shall be approved in writing by the Naval Surface Warfare Center, Indian Head, MD 20640-5035 prior to production.

3.4.2 Materials. All materials used in the manufacture of the impulse cartridge shall conform strictly to the specifications referenced on the respective drawings unless specific written approval outlining the departure has been obtained from the cognizant design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder’s selections shall be clearly stated in the proposal.

3.4.2.1 Smokeless powder: handling and loading. The smokeless powder shall be subjected to no unnecessary handling or exposure to the atmosphere. The handling and loading of smokeless powder exposed to the atmosphere shall be performed only under conditions that do not permit relative humidity to exceed 75 percent.

3.4.2.2 Smokeless powder, condition. The smokeless powder used shall exhibit no evidence of decomposition.

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3.4.3 Identical configuration. The impulse cartridge design and construction shall be identical to that subjected to qualification testing and approved for service use by the Naval Surface Warfare Center, Indian Head, MD 20640-5035.

3.4.4 Interchangeability. All impulse cartridges shall be functionally, structurally, and dimensionally interchangeable with respect to installation and performance.

3.5 Primary components. Only primary components from a single lot shall be used in a production lot of impulse cartridges. One primary component production lot may be used in more than one impulse cartridge production lot.

3.6 Impulse cartridge inspection.

3.6.1 Impulse cartridge production. The impulse cartridges shall be manufactured in accordance with drawing 3205AS372. Each impulse cartridge produced shall meet the requirements of the production inspections as outlined in Table I prior to being submitted for lot acceptance.

Table I. Production Inspection

Item	Nature of test	Requirement	Test method	Classification (DOD-STD-2101)
Output Material	Type and Weight	3205AS372	-	(C1)
Impulse cartridge	Serial number identification	3.2	4.2	(M101)
Primary Component	Single lot verification	3.5	4.5	(M102)
Impulse cartridge	Visual inspection	3.7.1	4.7.1	(M103)
Impulse cartridge	Radiographic examination	3.7.3	4.7.3	(M104)
Impulse cartridge	Bridge-wire resistance (BW)	3.7.4	4.7.4	(M105)
Impulse cartridge	Bridgewire Filter Test (BF)	3.7.5.	4.7.5	(M106)
Impulse cartridge	Electrostatic Discharge (ESD)	3.7.6	4.7.6	(M107)
Impulse cartridge	Leakage	3.7.5	4.7.5	(M108)

3.6.1.1 Inspection data. All data generated during Table I inspections shall be maintained at the contractor facility for two years beyond Navy shelf life.

3.6.2 Lot acceptance. Production lot acceptance shall be conducted in accordance with Table II. Failure of any impulse cartridge to meet the requirements of Table II shall result in rejection of the lot of impulse cartridges. The sample for ballistic performance shall be divided into thirds. For samples not divisible by three, the extra impulse cartridge will be tested at -65 °F. When two extra impulse cartridges are available, the second impulse cartridge will be tested at 160 °F. Acceptance of a impulse cartridge lot requiring a deviation or waiver shall have concurrence from the appropriate Naval Air Systems Command Headquarters program office if the impulse cartridge performance adversely reduces the system or subsystems factor of safety.

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Table II. Lot Acceptance Inspection

Nature of Test	Requirement	Test method	Classification (DOD-STD-2101)
Single lot verification	3.5	4.5	(M102)
Visual inspection	3.7.1	4.7.1	(M103)
Radiographic examination	3.7.3	4.7.3	(M104)
Bridge-wire resistance	3.7.4	4.7.4	(M105)
Filter Test	3.7.5.	4.7.5	(M106)
Electrostatic Discharge	3.7.6	4.7.6	(M107)
Leakage	3.7.7	4.7.7	(M108)
Ballistic performance (-65 °F, 70 °F, 160 °F)	3.11	4.11	(C2)
Preservation-packaging, packing and marking	5	-	(M109)

3.7 Non-Destructive Conditions.

3.7.1 Visual inspection. The impulse cartridge shall be of production standard and free of defects and have proper identification markings.

3.7.2 Receiver gage. The impulse cartridge shall pass the receiver gage test accordance with drawing 10001-929854C.

3.7.3 Radiographic examination. The impulse cartridge shall be free of defects or foreign material.

3.7.4 Bridgewire Resistance. The impulse cartridge shall have a bridge-wire circuit resistance of 0.90 to 1.10 ohms, when tested per 4.7.5.

3.7.5 Bridgewire Filter. The impulse cartridge shall exhibit 50 millivolts peak to peak maximum.

3.7.6 Electrostatic Discharge. The impulse cartridge shall not function or incur any internal/external damage or degradation as a result of the electrostatic discharge test.

3.7.7 Leakage. Each impulse cartridge shall meet the requirements of this section.

3.7.7.1 Production. The impulse cartridges built for fleet installation or first article test shall exhibit a leakage rate not greater than 1.0×10^{-5} cm^3/s of air at a pressure differential of 1 ± 0.1 atmosphere at the conclusion of production.

3.8 Baseline destructive conditions.

3.8.1 Low temperature conditioning. The impulse cartridge shall meet the performance conditions of section 3.11 when functioned after being conditioned at -65 ± 5 °F for a minimum of four hours.

3.8.2 Normal temperature conditioning. The impulse cartridge shall meet the performance conditions of section 3.11 when functioned after being conditioned at 70 ± 5 °F for a minimum of four hours.

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3.8.3 High temperature conditioning. The impulse cartridge shall meet the performance conditions of section 3.11 when functioned after being conditioned at 160 ± 5 °F for a minimum of four hours.

3.9 Environmental conditions.

3.9.1 Test program.

3.9.1.1 Temperature and humidity (T&H) cycling. The impulse cartridge shall not fire or incur defects or degradation as a result of the 14 day TH cycling test. The impulse cartridge shall also meet the performance conditions of section 3.11.

3.9.1.2 Salt fog. The impulse cartridge shall not incur detrimental degradation or damage during the salt fog test. The impulse cartridge shall also meet the performance conditions of section 3.11.

3.9.1.3 Shock. The impulse cartridge shall not fire or incur any defects or degradation as a result of 20g sawtooth pulses of 11-millisecond in duration. The impulse cartridge shall also meet the performance conditions of section 3.11.

3.9.1.3 Vibration. The impulse cartridge shall not fire or incur defects or degradation as a result of the random vibration and random vibration profiles. The impulse cartridge shall also meet the performance conditions of section 3.11.

3.10 Safety conditions.

3.10.1 Six-foot drop. The impulse cartridge shall not initiate or incur any defects or degradation after being dropped from a height of 6 feet +3, -0 inches onto a two-inch thick steel plate imbedded in concrete. The impulse cartridge shall also meet the performance conditions of section 3.11.

3.10.2 Forty-foot drop. The impulse cartridge shall not initiate or incur any internal/external defects or degradation that makes the impulse cartridge unsafe for disposal when dropped from a height of 40 feet +3, -0 inches onto a two inch thick steel plate embedded in concrete.

3.11 Performance condition.

3.11.1 Ballistic firings. The impulse cartridge when temperature conditioned at -65 °F, +70 °F or +160 ºF and functionally fired electrically in testing in a cable cutter, Mine Safety Appliance Company Part No. TA97116 or equal. A 3/16 inch-diameter steel cable in accordance with MIL-W-83420, Type I, Composition B, shall meet the complete severance of the steel cable within the 0.250 second.

3.11.2 Integrity. No impulse cartridge, its subassemblies, or individual part shall fragment, rupture, or fail causing a system failure or a marginal or unsafe condition.

3.11.3 Misfires. There shall be no misfires.

3.11.4 Retests. There shall be no re-tests.

3.12 Test failure. The impulse cartridge that does not meet any requirement of paragraphs 3.4 through 3.11 shall result in the failure of the entire lot.

3.13 Acceptance criteria. The impulse cartridges shall meet the requirements of 3.4 through 3.11. The acceptable number of defects is 0 and the reject number is 1.

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3.14 Workmanship. The impulse cartridges shall be constructed and finished in a manner to assure compliance with all requirements of this specification and associated drawings. The impulse cartridges shall be free of the following defects: burrs, dents, improper crimp, bent pins or other defects which may prevent entry of the impulse cartridge into the firing chamber or affect resistance to the moisture. All impulse cartridge making and identification shall be verified as a proper (See 5.7.1).

4. Verification

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.1.1)
b. Production inspection (see 4.1.2)
c. Lot acceptance inspection (see 4.1.3)

4.1.1 First article inspection. Unless otherwise specified in the contract or purchase order a first article sample of 69 impulse cartridges shall be prepared for inspection and test. This sample, manufactured using the same methods, processes and procedures proposed for the production lot, is for the purpose of determining that the contractor, prior to starting deliverable "fleet" production, is capable of producing impulse cartridges which comply with the technical requirements of the contract. A first article sample acceptable for being submitted to environmental and ballistic conditions shall be defined as a sample that has met the requirements of Table I. The contractor shall deliver the impulse cartridges to the activity designated by the contracting agency for test. First article requirements and verifications shall consist of the tests shown in Table III. Failure of any impulse cartridge to comply with the requirements of section 3 shall be cause for rejection of the first article lot. Any quantity production prior to notification by the contracting agency of the first article sample acceptability shall be at the contractor's risk. The first article impulse cartridges will not be applied as a part of the deliverable quantity specified by the contract.

4.1.2 Production inspection. Verification of all production impulse cartridges manufactured under the specification shall be inspected or tested, as required, for defects. Impulse cartridges failing to meet the requirements in Table I shall be rejected and removed from the lot.

4.1.2.1 Classification of characteristics. The characteristics, verified by inspection, demonstration, analysis, or test are classified as Critical, Major, or Minor in accordance with DOD-STD-2101. Tests and inspections that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the test or inspection. Tests and inspections that are not annotated with a classification code are classified minor.

4.1.3 Lot acceptance inspection. Verification for lot acceptance of the impulse cartridge lot shall be through inspection and test as specified in Table II. Failure of any impulse cartridge to comply with the requirements listed in Table II shall be cause for rejection of the lot presented. Packaging defects shall be corrected before acceptance of the production lot.

4.1.3.1 Lot sample selection. Inspection shall be used to verify the sample was randomly selected from the production lot. The sample will not be applied as a part of the quantity specified for delivery by the contract. Unless otherwise specified in the contract, the sample size shall be in accordance with ANSI/ASQ Z1.4, Inspection Level II, and as shown in Table IV. The production lot shall contain both the test and retained sample quantity.

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Table III. First Article Inspection

Test/Inspection	Reqmt. Para.	Verif. Para.		Quantity/Group									
			A	B	C	D	E	F	G	H	I	J
			6	6	6	12	12	12	12	6	6	9
Non-Destructive (Incoming)													
Visual Inspection	3.7.1	4.7.1	6	6	6	12	12	12	12	6	6	9
Radiographic examination	3.7.3	4.7.3	6	6	6	12	12	12	12	6	6	9
Bridgewire resistance	3.7.4	4.7.4	6	6	6	12	12	12	12	6	6	9
Filter Test	3.7.5.	4.7.5	6	6	6	12	12	12	12	6	6	9
Electrostatic Discharge	3.7.6	4.7.6	6	6	6	12	12	12	12	6	6	9
Leakage	3.7.7	4.7.7	6	6	6	12	12	12	12	6	6	9
Baseline – Destructive													
Ballistic Firing at -65 °F	3.8.1	4.8.1	6									
Ballistic Firing at 70 °F	3.8.2	4.8.2		6								
Ballistic Firing at 160 °F	3.8.3	4.8.3			6							
Environmental Conditions													
14 Day, TH Cycling	3.9.1.2	4.9.1.1				12						
Visual/Leak/BR/BF	3.7	4.7				12						
Salt Fog	3.9.1.1	4.9.1.1					12					
Visual/Leak/BR/BF	3.7	4.7					12					
20 G Shock	3.9.1.1	4.9.1.1						12				
Visual/X-ray/Leak/BR/BF	3.7	4.7						12				
Vibration (Random)	3.9.1.4	4.9.1.4.1							12			
Visual/X-ray/Leak/BR/BF	3.7	4.7							12			
Safety Conditions													
Six Foot Drop	3.10.1	4.10.1								6		
Visual/X-ray/Leak/BCR	3.7	4.7								6		
Forty Foot Drop	3.10.2	4.10.2									6	
Performance Conditions													
Ballistic Firing at -65 °F	3.11	4.11	6			4	4	4	4	2	2	
Ballistic Firing at 70 °F	3.11	4.11		6		4	4	4	4	2	2	
Ballistic Firing at 160 °F	3.11	4.11			6	4	4	4	4	2	2	
Spare Units	-	-										9

Table IV. Lot Sample Selection

Deliverable Production Lot Size	Test Sample Size	Retained Sample Size
51 to 90	13	3
91 to 150	20	3
151 to 280	32	3
281 to 500	50	3
501 to 1200	80	3
1201 to 3200	125	3
3201 to 10,000	200	3
10,001 to 35,000	315	3

4.1.3.2 Packing, packaging, and marking. Verification of the packing and packaging of the impulse cartridges and container markings conforming to Section 5 of this specification and the contract shall occur by inspection.

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4.2 Serial number. Verification shall occur by inspection of impulse cartridges and documentation.

4.3 Flight critical parts. Verification shall occur by inspection and documentation review.

4.4 Design and construction. Design and construction shall be verified by inspection of the technical drawing package, ammunition data cards, and by prototype demonstrations.

4.5 Primary components. Verification of single lot primary components shall be conducted by inspection of ammunition data cards.

4.6 Impulse cartridge inspection. Verification of both the impulse cartridge production and lot acceptance shall be done through test and the inspection of documentation. The individual verifications, as indicated in Table I and II, makeup these inspections which are specified within this document.

4.7 Non-destructive conditions.

4.7.1 Visual inspection. All impulse cartridges shall be verified by visual inspection and shall have no defects. Any impulse cartridge exhibiting defects shall be rejected. Identification markings, contract/drawing requirements, shall also be verified. Lot numbers shall be verified in accordance with MIL-STD-1168.

4.7.2 Receiver Gage. Each impulse cartridge shall be inserted in the receiver gauge test per drawing 100-928945C for verification. The impulse cartridge shall slide in the receiver gage with ease.

4.7.3 Radiographic examination. Each impulse cartridge shall be verified by radiographic examination in accordance with ASTM E 1742; any observable defects or degradation in assembly shall be cause for rejection. The impulse cartridge shall be positioned for the most revealing exposure with the long axis of each assembly perpendicular to the rays of the X-ray source. All impulse cartridges shall be identified with a serial number prior to examination. The impulse cartridge serial numbers shall be in consecutive order beginning with the lowest number in each production lot. The impulse cartridges shall be arranged on trays or boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the impulse cartridge displayed thereon. The radiograph identification shall include the complete lot number and the span of serial numbers displayed. Discontinuities in serial numbers shall be identified on the x-ray film. Setting on the radiographic equipment shall be noted on the radiographic review sheets.

4.7.4 Bridgewire resistance. The bridge circuit resistance of each of the two bridge wires shall be tested in accordance with MIL-STD-202, Method 303. The measurement shall be made at or corrected to 70°F. The resistance of the bridge circuit shall be measured with an accuracy of ± 1% using a current of less than 50 milliamps.

4.7.5 Bridgewire Filter Test. The signal generator shall be calibrated with a 1-ohm resistor for a load. A pulse of alternating current 100.0 ± 0.10 millivolts peak to peak, at a frequency of 1.0 ± 0.01 megahertz, shall be applied for 10 milliseconds maximum between the impulse cartridge pin and the body through a 47 ± 10% Ohm, 0.25 Watt minimum resistor (Figure 1). The voltage drop across the impulse cartridge studs shall be monitored using an oscilloscope set to the appropriate scale. The oscilloscope shall be capable of a minimum of 20 meg-samples per second. The oscilloscope lead shall have 50 Ohm terminator. With the CCU-167/B impulse cartridge installed in place of the calibrated 1 Ohm resistor, the voltage across the impulse cartridge shall meet the requirements 3.7.5.

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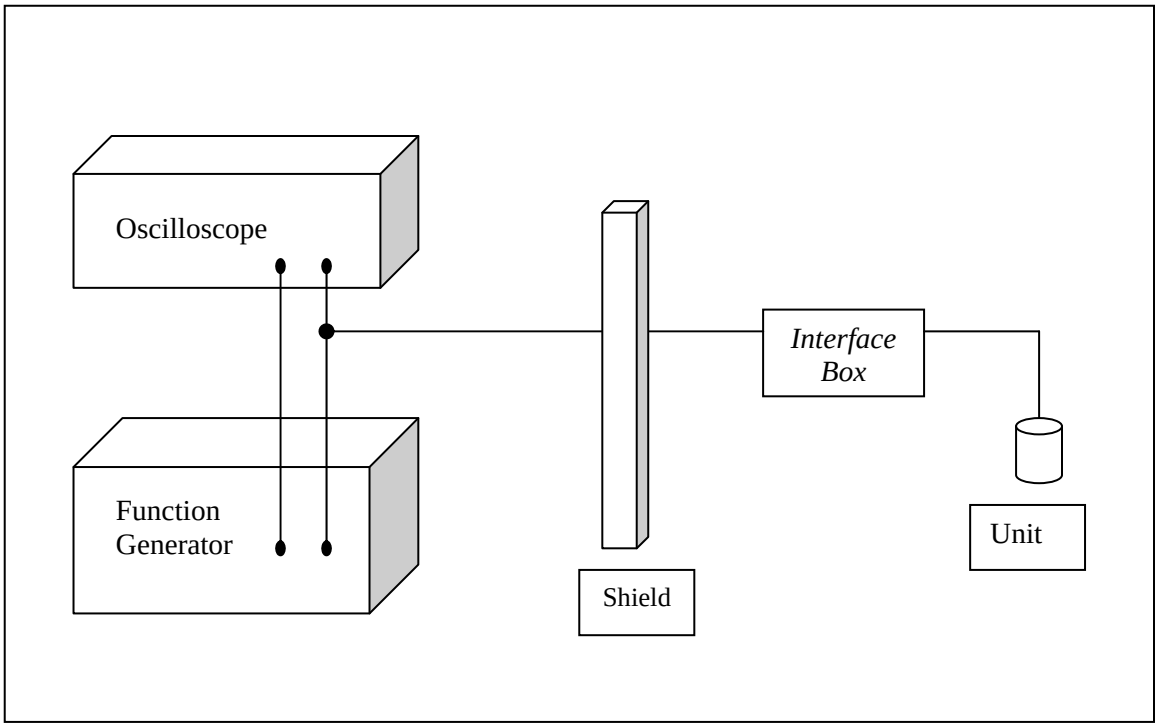


Figure 1: Bridgewire Filter Test Setup

4.7.6 Electrostatic Discharge (ESD). The impulse cartridge shall meets the requirements of 500 ± 5 percent picofarad capacitor charged to 25000 ± 500 volts and 5000 ± 5 percent and 500 ± 5 percent ohm resistor shall be connected in a 5 microhenry total inductance series circuit between pin and the case of the initiator.

All impulse cartridges shall be tested using the 5000 ± 5 percent ohm resistor and then using the 500 ± 5 percent ohm resistors. Switching in this circuit shall be accomplished by bringing together two 0.5 ± 0.05 inch spherical metal electrodes from an initial separation of 3 inches in air. Each series test shall constitute a separate test. Impulse cartridges used for this test shall be temperature conditioned for a minimum time of 12 hours at 70° ± 5 °F.

4.7.7 Leakage. All impulse cartridges in the production or first article lot shall be leak tested in an appropriate dry gas leak tester. The helium bombing time shall be for 60 minutes. After removal of the impulse cartridges from the bombing chamber they shall be washed in nitrogen or air. The leakage test will be performed immediately after washing using appropriate detection equipment. If a failing result is obtained, an appropriate detection probe will be used to determine at what location the device is leaking at this failing rate.

4.7.8 Salt Fog. The impulse cartridges shall be subjected to the salt fog test outlined in MIL-STD-810, Method 509.5 for 48 hours.

4.8 Baseline destructive conditions. Inspection and test shall verify the acceptability of the impulse cartridges for these first article tests and act as the baseline to measure the environmental and safety conditions against.

4.8.1 Low temperature conditioning. The impulse cartridges shall be verified by inspection and test by conditioning for a period of time not less than 4 hours and not greater than 24 hours at -65 °F. At the end of

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the conditioning, each device shall be visually inspected for degradation due to the conditioning environment. Each impulse cartridge shall be ballistically fired in accordance with 4.11.1.

4.8.2 Normal temperature conditioning. The impulse cartridges shall be verified by inspection and test by conditioning for a period of time greater than 4 hours at 70 °F. At the end of the conditioning, each device shall be visually inspected for degradation due to the conditioning environment. Each impulse cartridge shall be ballistically fired in accordance with 4.11.1

4.8.3 High temperature conditioning. The impulse cartridges shall be verified by inspection and test by conditioning for a period of time greater than 4 hours and not greater than 24 hours at 200 °F. At the end of the conditioning, each device shall be visually inspected for degradation due to the conditioning environment. Each impulse cartridge shall be ballistically fired in accordance with 4.11.1

4.9 Environmental conditions. After the below prescribed conditions for first article testing, a series of non-destructive tests (NDT) are to occur as outlined in Table III. The appropriate NDT requirements and verification methods are provided in Section 3.7 and 4.7.

4.9.1 Environmental test program.

4.9.1.1 Temperature & humidity cycling. The impulse cartridge verification shall be by inspection and test. The impulse cartridge shall be subject to the environment outlined in MIL-D-21625G (AS) with the below requirements.

a. The impulse cartridges shall be supported in such way that all areas are exposed to the prescribed atmospheric conditions at all times throughout the test. No impulse cartridge shall be touched by another impulse cartridge during the test.

b. The impulse cartridge shall be submitted to the following cycling schedule shown in Table V. This schedule shall be followed for a total of two (2) weeks (14 days) except that on the second weekend the soak time shall be from 1200 on Friday until 0800 Monday at a temperature of 160˚F at 95% relative humidity.

c. After the completion of the cycling test, each impulse cartridge shall be examined for evidence of mechanical failure or any other defects or degradation that may affect proper installation or functioning.

d. Following this condition and NDTs, the impulse cartridges are ballistically fired in accordance with 4.11.1 and as specified in Table III.

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TABLE V: Temperature Shock/Humidity/Altitude Cycling Schedule

Monday	
800	Place cartridges in a chamber maintained at +70 °F at 50% relative humidity (RH).
1200	Raise chamber temperature to +160 °F and the RH to 95%. The chamber temperature shall reach +160 °F at 95% RH no later than 1300.
1600	Remove cartridges from chamber and immediately place in a chamber maintained at -65 °F at a pressure altitude of 70,000 feet (0.65 psi).
Tuesday:	
800	Remove cartridges from chamber and immediately place in a chamber maintained at +70 °F at 50% RH.
1200	Remove cartridges from chamber and immediately place in a chamber maintained at –65 °F at a pressure altitude of 70,000 feet (0.65 psi}.
1600	Remove cartridges from chamber and immediately place in a chamber maintained at +160 °F at 95% RH.
Wednesday:	
0800	Reduce chamber temperature to +70 °F at 50% RH. The chamber temperature shall reach +70 °F conditions at 50% RH no later than 0900.
1200	Raise chamber temperature to +160 °F at 95% RH. The chamber temperature shall reach +160 °F at 95% RH no later than 1300.
1600	Remove cartridges from chamber and immediately place in chamber maintained at –65 °F at a pressure altitude of 70,000 feet (0.65 psi).
Thursday:	
0800	Remove cartridges from chamber and immediately place in a chamber maintained at +70 °F at 50% RH.
1200	Remove cartridges from chamber and immediately place in a chamber maintained at –65 °F at a pressure altitude of 70,000 feet (0.65 psi).
1600	Remove cartridges from chamber and immediately place in chamber maintained at +160 °F at 95% RH.
Friday:	
0800	Reduce chamber temperature to +70 °F at 50% RH. The chamber temperature shall reach +70 °F at 50% RH no later than 0900.
1200	Raise the chamber temperature to +160 °F at 95% RH. The chamber temperature shall reach +160 °F at 95% RH no later than 1300.
1600	Remove cartridges from chamber and immediately place in a chamber maintained at –65 °F at standard ambient pressure

This schedule shall be followed for a total of four (2) weeks (14 days) except that on the second and fourth weekends the soak time shall be from 1200 on Friday until 0800 Monday at a temperature of 160 °F at 95% RH.

4.9.1.2 Salt Fog. The impulse cartridges shall be subjected to the salt fog test outlined in MIL-STD-810, Method 509.5, for 48 hours with the following requirements:

a. Waxed string shall be used to suspend each impulse cartridge in the chamber.

b. Alternate 24-hour periods of salt fog exposure and drying conditions for a minimum of four 24-hour periods (two wet and two dry)

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c. Maintain the temperature in the exposure zone at 35 ±2°C (95 ±4°F).

d. Ensure the air velocity in test chambers is minimal (essentially zero).

e. Adjust the salt fog fallout such that each receptacle collects from 1 to 3 ml of solution per hour for each 80 cm^2 of horizontal collecting area (10 cm diameter).

f. During the entire exposure period measure the salt fog fallout rate and pH of the fallout solution at 12-hour intervals. Ensure the fallout is between 1 and 3 ml/80 cm^2/hr.

4.9.1.3 Twenty (20) G shock. The impulse cartridge verification shall be by inspection and test. The impulse cartridge shall be submitted to the shock test outlined in MIL-STD-810, method 516, procedure I with the below requirements.

a. The impulse cartridge shall be installed in test fixture F40 that tests 12 impulse cartridges at a time.

b. The shock pulse “wave” form shall be terminal peak saw-tooth Pulse. The peak amplitude shall be 20 g with duration of 11 milliseconds (Figure 1). Three shocks shall be applied in each direction of three mutually perpendicular axes. Each impulse cartridge shall be examined after each axis direction for evidence of mechanical failure or any other defects or degradation that may affect proper installation or functioning.

c. Following this condition and NDTs, the impulse cartridges are ballistically fired in accordance with 4.11.1 and as specified in Table IV.

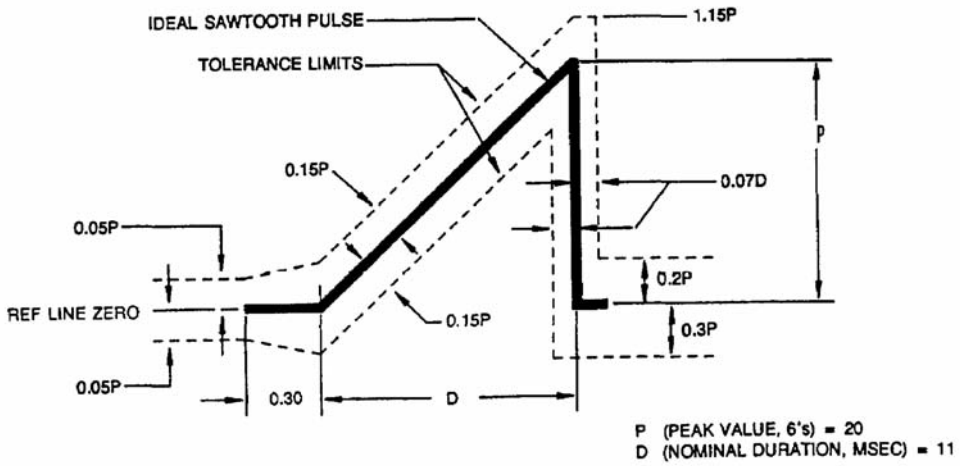


Figure 2: Terminal Peak Sawtooth Shock Pulse

4.9.1.4 Vibration. This test shall be conducted per Method 514.6 of MIL-STD-810, procedure I, shall be installed in the F40 test fixture and hands tighten the cap screws. The impulse cartridges shall be installed in vibration test fixture and temperature conditioned at 70 ±5°F

a. The impulse cartridges shall be subjected to random vibration along axis I for one hour. The vibration test levels are shown in Figure 2. After vibration along axis I. This procedure shall be repeated along axis II. After vibration along axis II, This procedure shall be repeated along axis III.

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b. The procedure outlined in paragraph 4.9.1.4 shall be repeated at –65°F ±5°F.

c. The procedure outlined in paragraph 4.9.1.4 shall be repeated at 200°F ±5°F.

d. After completion of random vibration test, each impulse cartridges shall be visually inspected for evidence of mechanical failure and any other defects or degradation that may affect proper installation or functioning.

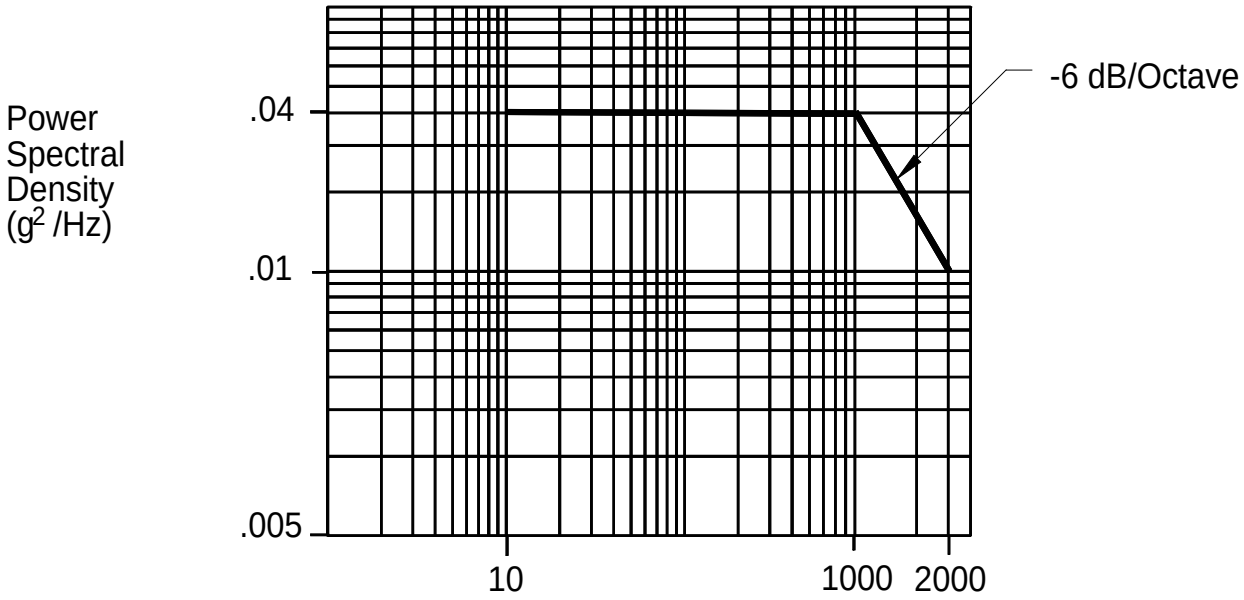


Figure 2: Vibration Test Levels

4.10 Safety tests. After the below prescribed conditions for first article testing, a series of non-destructive tests (NDT) occur as outlined in Table IV. The appropriate NDT requirements and verification methods are provided in Section 3.7 and 4.7.

4.10.1 Six-foot drop. Impulse cartridge verification shall be by inspection and test. The impulse cartridge shall be submitted to the six-foot drop test outlined in MIL-C-83125 with the below requirements.

a. The impulse cartridge shall be dropped without protective devices from a height of six feet onto a 2-inch thick steel plate supported by reinforced concrete. One new impulse cartridge shall be used for each drop orientation. The impulse cartridge shall be attached to a sliding guide by string or equivalent method. The guide shall properly lead the impulse cartridge to impact according to the required orientations.

Impact Orientation
Pin Down
Pin Up
Pin Horizontal

b. After the completion of the six-foot drop condition, each impulse cartridge shall be examined for evidence of mechanical failure, internally and externally, or any other defects or degradation that may affect proper installation or functioning.

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c. Following this condition and NDTs, the impulse cartridges are ballistically fired in accordance with 4.11.1as specified in Table III.

4.10.2 Forty-foot drop. Impulse cartridge verification shall be by inspection and test. The impulse cartridge shall be submitted to the forty-foot drop test outlined in MIL-STD-331, Test 103 with the below requirements.

a. The impulse cartridge shall be dropped without protective devices from a height of forty feet onto a 2-inch thick steel plate supported by reinforced concrete. One new impulse cartridge shall be used for each drop orientation. The impulse cartridge shall be attached to a sliding guide by string or equivalent method. The guide shall properly lead the impulse cartridge to impact according to the required orientations.

Impact Orientation
Pin Down
Pin Up
Pin Horizontal

b. After the completion of the forty-foot drop condition, each impulse cartridge shall be examined accordance with 4.7.1 for evidence of mechanical failure, internally and externally, or any other defects or degradation that may affect its ability of being disposed of safely.

4.11 Performance condition.

4.11.1 Ballistic firings. The impulse cartridges shall be verified by test following the non-destructive inspections designated in Table II. The impulse cartridges are ballistically fired within five minutes at the temperatures specified in Table III as applicable with the below requirements.

Each impulse cartridge shall be temperature conditioned for not less than 4 hours and no more than 24 hours at its specified temperature. The impulse cartridges shall be removed from the conditioning chamber and test fired within 5 minutes. If any impulse cartridge is not fired within 3 minutes after removal, it shall be reconditioned at the specified temperature for an additional 1 hour and then re-tested.

4.11.2 Integrity. The impulse cartridge shall be verified by inspection and analysis for structural and performance integrity. A marginality of success analysis shall occur by performing a disassembly, sectioning, or other method to confirm every component within the design has maintained structural integrity. A review of the ballistic firing data shall also be assessed by conducting statistical analysis for performance integrity.

4.11.3 Misfires. Verification shall occur by inspection of the impulse cartridge and performance firing documentation.

4.11.4 Retest. Verification shall occur by documentation and data inspection.

4.12 Test failure. Verification shall occur by inspection, demonstration, analysis, or test depending upon the specific verification in paragraphs 4.4 through 4.11. If a failure to meet an explicit requirement is attributable to a specific cause, excluding the impulse cartridge, the original results will be discarded and the verification conducted again.

4.13 Acceptance criteria. Verification shall occur by inspection, demonstration, analysis, or test depending upon the specific verification in paragraphs 4.4 through 4.11.

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4.14 Workmanship. Construction and finishing of impulse cartridge shall be verified by inspection, pre and post- Table II results.

5. PACKAGING.

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The CCU-167/B, impulse cartridge is used to actuate a helicopter-rescue hoist cable cutter (MSA Part No. TA 97116 or 97999, or equal) to jettison a load in an emergency. Since the impulse cartridge was developed for use on military helicopters application. This impulse cartridge was designed for military use and has no commercial applications.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. If required, the specific issue of individual documents referenced (see 2.2.1, 2.2.2 and 2.3).

c. Whether a first article inspection is required and if so, specify the test activity (see 3.1).

d. Markings if other than as specified (see 5.1 and 6.5.3).

e. If bar coding is not required on outer container (see 5.1).

f. Inspection conditions if other than specified in 4.1.

g. Production lot size (quantity) and test facility.

h. That the safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DOD 4145.26M are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition Ashore", OP 5 are applicable.

6.3 First article. When a first article inspection is required, the contracting officer should provide specific guidance to bidders whether the item(s) should be a pre-production sample, a first article sample, a first production item, a sample selected from the first production items, standard production item from the contractor's current inventory (see 3.1) and the number of items to be tested as specified in 4.1.3.1. The contracting officer should also include specific instructions in acquisition documents of the first articles. Invitation(s) for bid(s) should provide that the U.S. Government reserves the right to waive the requirements

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for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the U. S. Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior U.S. Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Data. For the information of contractors and contracting officers, any of the data specified in (a), subparagraphs below, (b) applicable documents listed in section 2 of this specification or (c) referenced lower-tier documents need not be prepared for the Government unless specified in the contract or order.

6.4.1 Standard deliverables. In addition to the delivery of impulse cartridges, the contract may require delivery of the following:

a. Radiographic review report
b. Radiographic LAT sample plates and technique
c. Requests for waiver/deviation and engineering change proposal (ECP)
d. Progress and/or management reports
e. Test and inspection report if contractor tested
f. Ammunition data card
g. Marriage log
h. Performance oriented packaging (POP) test report
i. Department of Transportation Hazard Classification

6.5 Contract packaging and marking. The following packaging and marking requirements should be specified in the contract.

6.5.1 Preservation and packing. Level A and commercial preservation/packing should be in accordance with the Code of Federal Regulation 49 CFR 171-179. The contractor is to be responsible for performance oriented packing certification and documentation in accordance with Regulation 49 CFR 171-179.

6.5.1.1 Packaging for service use. Level A preservation and packing should be used for packaging of all production lots for service use, design certification, first article, and production lot acceptance inspections. Impulse cartridges should be packaged one to every hermetically sealed inner container. The number of inner containers and the method used to pack the outer container should be as approved, based on the result of the accepted hazard classification testing. For an explanation of terms see 6.5.2.

6.5.2 Explanation of terms. For the purpose of this specification, NAVSUPINST 4030.28 (20 June 1975) paragraphs 1-4, 2-2, 203, and Appendix A, when applicable, is used to clarify terms.

6.5.3 Markings.

6.5.3.1 Normal markings. The Special Packaging Instructions (SPI) in accordance with the requirements stated herein will define all packing requirements. The specified marking information should be applied to the containers in accordance with the applicable provisions of MIL-STD-129. Lot number should be in accordance with MIL-STD-1168.

(1) NSN: 1377-01-550-8543
(2) DODIC: DWGY
(3) Impulse, Cartridge, CCU-167/B
(4) Drawing 3205AS372
(5) Quantity

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(6) Gross weight and cube
(7) Lot number in accordance with MIL-STD-1168
(8) Contract/purchase order number

6.6 Definitions.

6.6.1 Hangfire. A hangfire is defined as a firing in which the elapsed time between actuation of the firing mechanism and complete severance of the steel cable exceeds 0.250 second.

6.6.2 Flight critical part. A part, the single failure of which, during an operating condition, could cause loss of aircraft or one of its major components, loss of control, unintentional release of, or inability to release, or which may cause significant personnel injury, including in flight, escape, survival, or rescue.

6.6.3 Analysis. Verification by technical or mathematical models or simulation, algorithms, charts, graphs, or circuit diagrams, and representative data.

6.6.4 Defect. For the purpose of inspection or verification, the term defect is defined to mean burrs, dents, scratches, split or cracked edges, sharp edges, and all other defects which may prevent the installation, adversely affect the ballistic performance, or in any way adversely affect fit, form, function, safety or performance of the initiator.

6.6.5 Degrade. For the purpose of inspection or verification, the term degrade (degradation) is defined to mean chemical decomposition, fracture, yielding, deboning, delaminating, cracking, corrosion or any other condition which could adversely affect the form, fit, function or safety of this component

6.6.6 Demonstration. Verification by operation, adjustment or configuration of items performing their designed functions under the specific conditions. The items may be instrumented and quantitative limits or performance monitored, but only check sheets rather than actual data are required to be recorded.

6.6.7 Inspection. Verification by visual inspection of the item, reviewing descriptive documentation, and comparing the appropriate characteristics with predetermined standards to determine conformance to requirements without the use of special laboratory equipment or procedures.

6.6.8 Test. Verification through systematic exercising of the item under all appropriate conditions. Performance is quantitatively measured either during or after the controlled application of either real or simulated functional or environmental stimuli. The analysis of data derived from a test is an integral part of the test and may involve automated data reduction to produce the necessary result.

6.6.9 Misfire. Failure to meet any ballistic performance condition will be deemed a misfire.

6.7 Subject term (keyword) listing.

Cable Cutter
Bolt Cutter
JM96
Wire Rope

Custodian:
Navy – OS

Preparing activity:
Navy - OS
(Project 1377-2011-001)

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